Durham-Jazan at Daleel 2026: Domain-Routed Ensembles, LLM Augmentation, and the Limits of Decision-Rule Tuning for Arabic Argument Mining

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Abstract

We describe the Durham-Jazan submissions to the Daleel 2026 shared task on Arabic argumentative discourse mining, which comprises multi-label paragraph classification (Task 1) and argumentative span detection (Task 2), each with a Closed and an Open track. Our Task 2 Closed system routes the Modern Standard Arabic editorials to CAMeLBERT and the noisier debate transcripts to MARBERTv2. The Open system blends the two families at character level and adds synthetic training data from a large language model for rare-class generalisation, generated by a segment-concatenation recipe that makes span offsets exact by construction. The Task 1 systems add back-translation, targeted augmentation of the rare classes, and a QLoRA-tuned Qwen3-32B for the editorials. Everything else trains on one 16 GB consumer GPU. Our official test scores are 0.729 and 0.736 overlap-F1 on Task 2 and 0.657 and 0.669 macro-F1 on Task 1 (Closed and Open). We placed first of two on Task 1 Open, first of one on Task 2 Open, fifth of ten on Task 1 Closed, and sixth of eight on Task 2 Closed. Our central finding concerns our own systems on this data. Recalibrating per-class thresholds on validation outputs overfits them. Under a controlled protocol the held-out effect is negative at every tuning-set size we tested, while the apparent in-sample gain shrinks as the tuning set grows. An audit of eighteen interventions from our campaign is consistent with this, though observational. A size-matched study links the editorial deficit that every team showed to the genre composition of the training data. We release our code and synthetic data.[1]

1 Introduction

Argument mining, the automatic identification of argumentative structure and strategies in text (Lawrence and Reed, 2019), remains under-resourced for Arabic. The Daleel 2026 shared task (Nabhani et al., 2026) addresses this with two subtasks over Arabic news editorials and debate transcripts. **Task 1** asks for multi-label classification of paragraphs into six argumentative categories and **Task 2** for character-level detection of labelled argumentative spans. Each subtask has a Closed track (public models and the provided data only) and an Open track (any resources, including proprietary large language models).

Team **Durham-Jazan** took part in all four tracks. On Task 2 Closed we routed the Modern Standard Arabic (MSA) editorials to CAMeLBERT-mix (Inoue et al., 2021) and the debates to the dialect-pretrained MARBERTv2 (Abdul-Mageed et al., 2021), and on Task 2 Open we blended the two families. We target the dominant error source, rare-class generalisation, with task-adaptive pretraining (Gururangan et al., 2020), back-translation (Sennrich et al., 2016), and synthetic data from a large language model, and, for Task 1, route the editorials to a Qwen3-32B (Yang et al., 2025) fine-tuned with QLoRA (Dettmers et al., 2023).

Our best official test scores are 0.729 and 0.736 overlap-F1 on Task 2 (Closed and Open) and 0.657 and 0.669 macro-F1 on Task 1, placing first on both Open tracks, fifth on Task 1 Closed, and sixth on Task 2 Closed (Table 1). Beyond the systems, we report two methods and two findings. The first is *segment-concatenation generation*, which makes LLM-generated span annotations exact by construction for ordered, non-overlapping spans (Section 3.3). The second is *metric-aware span compaction*, worth 0.017 on test at no GPU cost (Section 3.1). The third is a controlled experiment showing that, for our systems on this data, recalibrating thresholds on validation outputs overfits them at every tuning-set size we tested (Appendix C), with an observational audit of eighteen interventions from our campaign pointing the same

[1]https://github.com/alshehriamal1994/daleel2026-durham-jazan

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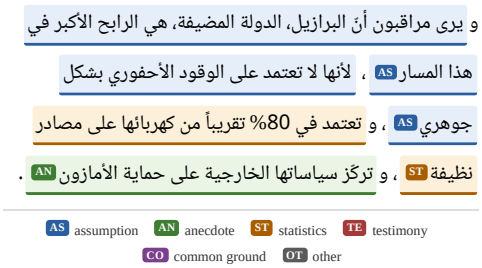


Figure 1: Gold Task 2 annotation of a development-set editorial paragraph, with the label key for the whole paper. Connectives such as *wa* ("and") fall outside the spans. The paragraph translates as *"Observers see host-country Brazil as the biggest winner, since it does not fundamentally depend on fossil fuels, derives about 80% of its electricity from clean sources, and centres its foreign policy on protecting the Amazon."* The Task 1 target is {AS, ST, AN}.

way (Figure 4). The fourth is a size-matched study linking the editorial deficit to training-data genre (Table 11, with the label-level pattern behind it in Figure 5).

2 Background

Task setup. The corpus contains paragraphs from Arabic news editorials and televised debates. In Task 1 the output for a paragraph is a subset of six evidence types, namely assumption (AS), anecdote (AN), statistics (ST), testimony (TE), common ground (CO), and other (OT), in the argumentation-strategy tradition of Al-Khatib et al. (2016). Task 2 asks for labelled character spans, and Figure 1 shows a worked gold example of both. Task 1 is scored with macro-F1 over the six labels. Task 2 uses an overlap-based F1 in which a predicted span p accrues $|p \cap g|/|p|$ over every same-label gold span g it overlaps and g symmetrically accrues $|p \cap g|/|g|$. Precision is normalised by the predicted span count and recall by the gold count.

Data. The data comprise 612 training, 217 development, and 213 test paragraphs (87 editorial and 126 debate), with the dev gold released for the evaluation phase. In the 612 training paragraphs the Task 1 label support is AS 462, OT 222, AN 182, TE 160, CO 36, and ST 28, and those same paragraphs carry 2,975 Task 2 spans, 4.86 per paragraph. Label distributions differ sharply by domain (editorials ST 22 and OT 20 against debates ST 6 and OT 202), motivating domain routing. In all tracks we trained on both domains, as declared to the organisers.

3 System Overview

3.1 Task 2: Span Detection

Base tagger. The encoder output passes through dropout and a linear layer with six per-token sigmoid units, and we train the tagger with clipped token-level class weighting (Appendix A). Spans are decoded as maximal token runs above per-class thresholds tuned against the official scorer, and mapped to characters through the tokeniser offset mapping. Sigmoid tagging, unlike BIO, can represent nested gold spans, and dev holds one (Figure 6).

Domain-routed ensemble. On the dev leaderboard the families were complementary per domain (0.664/0.699 for CAMeLBERT-mix against 0.626/0.715 for MARBERTv2), and our routed entry scored 0.706 (Appendix A, with a worked case in Figure 7). That entry also widened both ensembles, so it does not isolate routing. Three saved development decodes do, since the routed decode reuses the CAMeLBERT-mix editorial rows and the MARBERTv2 debate rows unchanged, and routing is worth 0.005 there over the stronger family alone. Encoder-family routing helped Task 2 only, and lost on Task 1 under the pooled macro metric. The final Closed system therefore routes each paragraph by its provided genre label, sending editorials to the mean of eight CAMeLBERT-mix seeds and debates to eight MARBERTv2 seeds, with domain-tuned thresholds. In the evaluation phase the encoders were further pretrained on the task text and trained on the combined train and dev gold (Section 4 and Figure 2). We label them DAPT-v2 throughout, the name used in our code, although the procedure is task-adaptive.

Metric-aware span compaction. Token-threshold decoding fragments long spans, and the partial-overlap metric counts every fragment in the precision denominator. Our post-processing merges same-label spans separated by at most G characters and drops spans shorter than L (G_{ed}=400, G_{db}=5, L=25). The gap and length values were grid-tuned on the dev gold by worst-case improvement across our Closed and Open routed development decodes, and the post-processing added 0.014 on test and 0.017 after a refined grid.

Open track. The Open system adds 158 synthetic span paragraphs (Section 3.3) and a character-

level probability blend across the two families (Appendix A). We rejected blending in the development phase because it fragmented spans at whitespace, and compaction removes that failure mode, reviving the blend (+0.005 on test, +0.012 on editorials). That submission also retuned the per-domain thresholds, which supply most of the out-of-fold editorial gain.

3.2 Task 1: Paragraph Classification

Encoder system (development phase). We use CAMeLBERT-mix with a six-way sigmoid head, a five-seed ensemble, and robust-median per-class thresholds over seeds. Three additions were validated on clean held-out splits (Section 4). Continued pretraining on the task paragraphs (task-adaptive pretraining, Gururangan et al., 2020) added 0.012, and back-translation through English with beam decoding on OPUS-MT added 0.040 (Tiedemann and Thottingal, 2020; Edunov et al., 2018). Targeted rare-class augmentation, two extra sampled back-translations for every ST or CO paragraph and three in total with the global pass, added a further 0.012. The amount of augmentation matters, as an eightfold version broke threshold calibration (dev 0.627 against 0.673).

Fine-tuned LLM routing (evaluation phase). The Closed track permits open-weight models up to 70B, fine-tuning included. We fine-tune Qwen3-32B with 4-bit QLoRA (r=16, loss on answer tokens only, Appendix A) as a generative classifier. On a controlled holdout it was the first model we tried that beat the tuned encoder ensemble (standalone seed mean 0.744, best 0.767, against 0.715), though with high seed variance. The final system therefore takes the union of three seeds' label sets on the editorials, a recall-oriented choice that entered without a separate validation, and keeps the encoder ensemble on the debates. A post-submission check on the same holdout supports it, with the union above its own seed mean in every three-of-four combination and above majority voting and intersection (Table 4). A one-seed routed submission added 0.008 to 0.011 on test, and the three-seed union a further 0.017 to 0.022. The Open variant changes only the debates, since both tracks route the editorials to the same Qwen3-32B union and return an identical official editorial score. It adds 291 synthetic paragraphs to the encoder's training data and span fusion from Task 2, uses per-domain per-class thresholds in place of the global out-of-fold recalibrated set, and keeps the single-family CAMeLBERT-mix ensemble where Closed blends in MARBERTv2 (Table 3).

Span fusion from Task 2 (Open). Paragraph labels can also be read off the Task 2 decode (a label counts as present when a compacted span of at least 25 characters carries it). Span-derived labels alone reached 0.727 out-of-fold macro-F1, against 0.725 for the deployed classifier. Fused with the classifier through per-class and per-domain rules, they lost on Closed and won on Open (Section 5.3).

3.3 Synthetic Data from an LLM (Open)

A proprietary large language model generated synthetic training paragraphs from few-shot prompts grounded in the official label definitions and real seed examples, targeting the rarest classes (ST, CO, editorial OT) and varying topic and register. Appendix B gives the protocol and two blind quality checks, one by a second model and one by a human annotator. For Task 1 we obtained 291 validated paragraphs in the same multi-label scheme. For Task 2 the model writes each paragraph as an ordered sequence of labelled segments, and joining them with single spaces yields exact character offsets by construction, a recipe we call *segment-concatenation generation*. Unlike inline markup, it needs no tag parsing or alignment, and it adapts generate-from-structure synthesis (Ding et al., 2020; Josifoski et al., 2023) to character-offset spans. We obtained 158 paragraphs with 772 spans at the real span density, boosting ST to 128 spans against 38. Synthetic data are used in training only, and threshold tuning and held-out evaluation always use real data. A larger pool did not help, as a 238-paragraph version gained on editorials and lost on debates (net -0.001).

4 Experimental Setup

Splits. The development phase used 5-fold cross-validation plus a clean held-out simulation (train on 395, predict 217 unseen), with dev scores from the Codabench leaderboard. The evaluation phase retrained on the 829 combined train and dev gold paragraphs, and local re-scoring of our saved dev submissions against the released gold reproduced all four leaderboard numbers exactly.

Models and preprocessing. We run CAMeLBERT-mix, MARBERTv2, and AraBERTv2 (Antoun et al., 2020) through

Track	Score	Ed	Db	Rank	Win.
Task 2 Open	0.736	0.627	0.766	1st	0.736
Task 2 Closed	0.729	0.610	0.762	6th	0.755
Task 1 Open	0.669	0.501	0.762	1st	0.669
Task 1 Closed	0.657	0.501	0.741	5th	0.712

Table 1: Final official test results. Ed and Db are the editorial and debate columns, and Win. the winning score, all taken from the organisers' official sheet. Task 1 drew ten Closed and two Open entrants, and Task 2 eight Closed and one Open, so we were the only team on Task 2 Open and the winning score in both Open rows is our own.

Transformers (Wolf et al., 2020), with maximum length 384 and fp16. DAPT-v2 continues masked language modelling for 30 epochs on all 1,042 task paragraphs (train, dev, and the unlabelled test text) in a transductive setting where labels are never used. Qwen3-32B uses 4-bit QLoRA with r=16. Per-task hyperparameters and library versions are in Appendix A.

Compute. One RTX 4080 SUPER (16 GB) ran all encoder work and models up to 14B parameters. One A100 80 GB at the Durham NCC ran Qwen3-32B for about six GPU hours.

5 Results

5.1 Official Results

Table 1 shows our final official test results, and Table 3 lists the components of each submission. At the close of the development phase we stood first on both Open boards and second and third on the Closed boards. The evaluation phase stayed close on the Closed boards, and our Task 1 Open score rose from 0.622 to 0.669 over four submissions. On Task 1 Open we lead the one other entrant by 0.078 overall, by 0.103 on editorials and by 0.110 on debates, and on Task 2 Open no other team entered.

5.2 Ablation Analysis

Table 2 breaks the Task 1 dev results down by class. The successive additions of DAPT, back-translation, rare-class augmentation, and synthetic data (holdout macros 0.644, 0.684, 0.696, and 0.723 from a 0.632 base, the synthetic step positive on all four holdout splits) chiefly lifted the classes they targeted. ST rises from 0.29 to 0.89 while the common classes barely move, and CO reaches only 0.21. On Task 2, dev gold with DAPT-v2 and wider ensembles added 0.014 on test (0.698 to 0.712),

T1 system	AS	AN	ST	TE	CO	OT	Mac.
pos_weight	.91	.54	.43	.70	.00	.70	.547
plain ens.	.93	.61	.29	.75	.00	.70	.546
DAPT + BT	.92	.63	.67	.79	.17	.69	.642
+ rare-aug 3×	.90	.65	.82	.84	.15	.67	**.673**
+ LLM-synth	.93	.64	.89	.81	.21	.67	**.691**

Table 2: Task 1 per-class dev F1, with every macro matching the development leaderboard exactly. BT is back-translation, and Mac. the macro-F1. The first two rows ensemble CAMeLBERT-mix with AraBERTv2, while the last three use DAPT-CAMeLBERT-mix with three, five and five seeds. The steps are therefore successive development submissions rather than single-factor ablations, and the controlled ladder is the clean holdout quoted in the text. Bold marks the two entries that led their development boards when they were submitted, the Closed board at .673 and the Open board at .691.

compaction another 0.014 and 0.003 more from a refined grid (0.729), and synthetic spans with the blend 0.007 on Open (0.736). A post-submission paired bootstrap over the development paragraphs puts compaction on a routed development configuration at +0.027 (95% interval [+0.014, +0.040], positive in all 2,000 resamples). Those parameters were tuned on the same paragraphs, so the interval is in-sample and not comparable to the test deltas above.

5.3 What Held Up, and What Did Not

Our central result concerns one intervention in our own pipeline, measured on this data under one protocol, and we claim it for that setting only. Recalibrating per-class thresholds on out-of-fold predictions overfits those predictions at every tuning-set size we tested. On a quarter of the out-of-fold data the recalibration appears to gain 0.033 in sample and loses 0.016 out of sample. Enlarging the tuning set to nine tenths shrinks the apparent gain to 0.010 while the held-out loss remains at 0.015 (Table 7). Tuning-set size changes the apparent gain, not the delivery. The result governs recalibration of an existing threshold set, not the presence of per-class thresholds, which every submitted configuration uses. Its comparison baseline is not fully blind, as Appendix C details.

The campaign audit is consistent with this but is observational and weaker. Model changes, meaning changes to data, encoders, or ensembles, gained in eight cases of twelve, against two of six for the decision rules, which are layers tuned on validation outputs. Validation overstated the delivered gain by

0.5 points on average for the former and 2.5 for the latter (Figure 4, Table 6). We attach no significance test, since the set records what we tried rather than a designed sample. The decision rules were also fitted on the predictions they are then scored against, while the model changes were judged on held-out data. That asymmetry runs in the direction of the contrast, so we treat the audit as context rather than evidence. Appendix C discusses a candidate explanation and the one entry that appears to contradict the controlled result.

5.4 Error Analysis

Every one of the twenty-one ranked entries on the four leaderboards scored lower on editorials than on debates. Our 0.610 is the lowest editorial column of the eight Task 2 Closed teams, against a best of 0.684 that belongs to a team other than the overall winner. Our own editorial column also fell from development to test on all four tracks, by 0.042 to 0.155, while our debate column rose by 0.040 to 0.075. Those four pairs compare different systems, since we rebuilt every track between the phases. Our only fixed-system comparison points the same way, as the development Task 2 Closed system re-run on test fell from 0.668 to 0.605 on editorials and rose from 0.722 to 0.739 on debates. Size-matched training reproduces the editorial-versus-debate gap from training composition alone (Appendix D). It does not account for our own development-to-test editorial drop, which we cannot explain from the data available to us. The organisers' sheet reports test scores only, so we cannot tell whether other teams saw the same drop. The leading Task 1 Closed entry, Salah Abdo at 0.712, beats us by 0.018 in each domain yet by 0.055 pooled. The pooled macro is computed over all 213 paragraphs rather than averaged over the two domain macros, so a gap of that shape has to come from labels that are rare within a domain, although we cannot confirm this without that team's per-label predictions.

6 Conclusion

We presented domain-routed and cross-family blended span ensembles with metric-aware compaction and, for Task 1, a hybrid of DAPT encoders and Qwen3-32B. Three things outlast the systems. Segment-concatenation generation makes LLM-generated span annotations exact by construction, and needs no alignment step. For our systems on this data, recalibrating thresholds on validation outputs overfits them at every tuning-set size we tested, under a controlled protocol whose limits we state. And the editorial deficit that every team showed follows the genre composition of the training data, which points to genre-balanced annotation rather than further modelling as the way to close it.

Limitations

Because DAPT-v2 is trained transductively on the unlabelled test paragraphs, the submitted pipeline cannot be applied unchanged to genuinely new text. The development-phase variant, trained without test text, is the inductive alternative. The evaluation phase allowed unlimited submissions. Most interventions were validated before submission. The exceptions are the final three-seed union, checked on the holdout only after submission (Table 4), and the configurations noted in Table 6. The choice among submitted variants used the shared leaderboard. Most test deltas in Table 6 come from single paired comparisons on 213 paragraphs. They are smaller than the unpaired bootstrap half-widths (Efron and Tibshirani, 1993) of the development set ($\pm$0.051 for Task 1 macro-F1 and $\pm$0.030 for Task 2), so we lean on the aggregate sign pattern. The same noise applies to rankings, including ours. The synthetic data come from a single proprietary generator whose vendor and version we withhold. A blind re-annotation of all 291 Task 1 paragraphs by a second model agrees with it at micro-F1 0.94, and a blind human check of 30 paragraphs at 0.84, with the targeted rare classes near-perfect (Appendix B). Segment concatenation cannot express nested spans (Figure 6). Appendices F and E give worked error examples and suggestions for the organisers. Finally, our editorial columns trail those of the winning systems on the two Closed tracks by 0.02 to 0.06. Appendix D traces most of the genre gap to the genre skew of the training spans, but how to close it remains open.

Ethics Statement

The synthetic training data released with this paper were generated throughout by a proprietary large language model. The statistics that appear inside synthetic evidence spans are plausible in form but invented. The repository documents these files as machine-generated, and they are suitable for training and methodological replication rather than as a source of factual claims. Synthetic data are

used in training only, and threshold tuning and held-out evaluation use the organisers' real data. The quality checks in Appendix B were run by a second large language model from the same family as the generator, which bounds annotation quality from above, and by the first author on a blind sample of 30 paragraphs. No wider human validation was performed.

The synthetic data are released under CC-BY-4.0 and the code under the MIT licence. The shared-task corpora are used under the organisers' terms and are not redistributed. The debate and editorial material concerns matters of public argument and contains no personal data beyond what the organisers released. Our evaluation-phase encoders continue masked language modelling on the unlabelled test paragraphs, a transductive setting that we describe in the Limitations, so systems intended for genuinely unseen text should use the inductive variant.

Argument mining systems of this kind identify what functions as evidence in a text. They do not assess whether that evidence is true, and should not be presented to readers as doing so.

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A Hyperparameters, Software, and Prompts

Encoders. CAMeLBERT-mix (`CAMeL-Lab/bert-base-arabic-camelbert-mix`), MARBERTv2 (`UBC-NLP/MARBERTv2`), and AraBERTv2 (`aubmindlab/bert-base-arabertv2`). PyTorch 2.10, Transformers 5.5, fp16, maximum length 384. All training uses AdamW, in its 8-bit paged variant for Qwen3-32B, through the Transformers Trainer for Task 1 and a hand-written loop for Task 2.

Task 2 tagger. 10 epochs, learning rate 3e-5, batch 8, dropout 0.1, and per-class token-level $\text{pos_weight} = \text{clip}(\sqrt{n_{\text{neg}}/n_{\text{pos}}}, 1, 8)$. Per-class decoding thresholds are tuned per domain by coordinate ascent on the official scorer. The thresholds of the final Closed decode (AS/AN/ST/TE/CO/OT) are [.6, .4, .55, .5, .3, .7] on editorials and [.45, .75, .8, .75, .8, .55] on debates, and the Open blend uses its own retuned set. The ensembles use seeds 42, 1, 2, 3, 4, 5, 6, and 7 for both encoder families. All per-system threshold configurations are released with the code. Final compaction uses G_{ed}=400, G_{db}=5, and minimum length 25, from a grid on the dev gold whose surface plateaus for $G_{\text{ed}} \geq 200$. The plateau means the overlap metric favours merging essentially all same-label spans within an editorial paragraph, a property future scorers may wish to revisit. On the development leaderboard, routing scored 0.706 against 0.689 for the best single-family CAMeLBERT-mix submission and 0.687 for MARBERTv2 (per-domain editorials and debates, 0.664/0.699 against 0.626/0.715). Those three entries are not matched, since the routed one averaged five CAMeLBERT-mix seeds on the editorials and three MARBERTv2 seeds on the debates, while the single-family entries used one CAMeLBERT-mix seed and a two-seed MARBERTv2 mean. Three matched development decodes are released instead, the routed one taking its editorial rows from the CAMeLBERT-mix decode and its debate rows from the MARBERTv2 decode. Under the official scorer they reach 0.670, 0.687, and 0.692, so routing at matched components is worth 0.005 over the stronger family. The rest of the leaderboard difference is ensemble width and per-domain threshold tuning, which we cannot separate because no development probabilities were saved for the wider runs. The Open char-level blend projects each family's token probabilities to characters, taking the maximum over the tokens covering a character, and averages the families with w_{CAM}=0.5 on editorials and 0.35 on debates.

Seed variance and ensemble size. In a post-submission check we trained eight seeds of the Task 1 encoder on the full training set and evaluated them on dev with thresholds fixed at 0.5, so that no calibration enters the comparison. These are plain encoders without augmentation or DAPT, so the absolute values sit below those of our submitted system. Single seeds range from 0.468 to 0.552 macro-F1, a spread of 0.084 that is wider than most

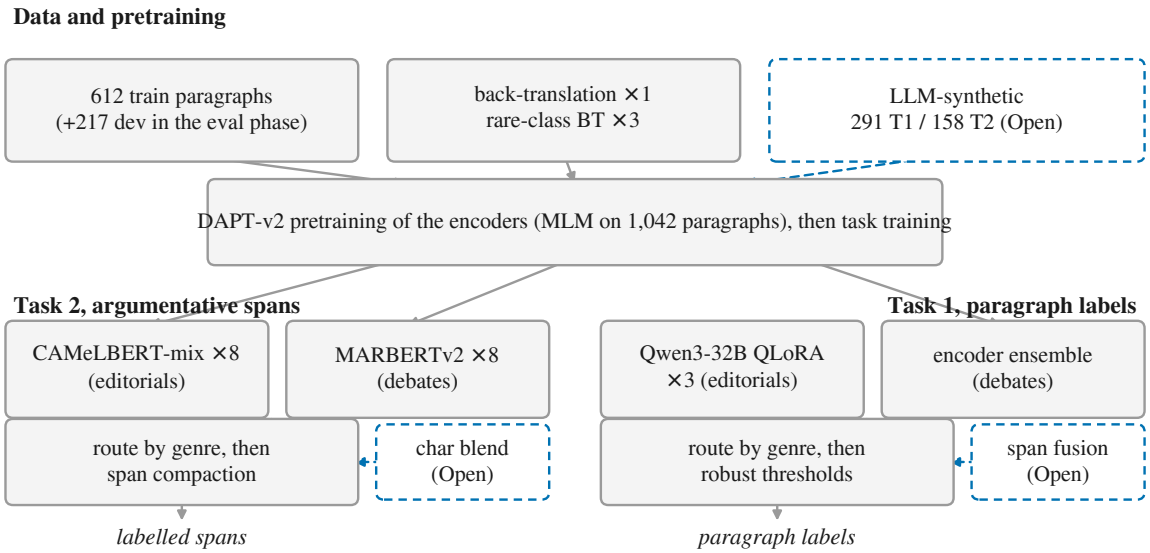


Figure 2: The four systems at a glance. Solid components form the Closed-track pipelines, and dashed blue components are the Open-track additions. Back-translation applies to Task 1 only, as it does not preserve offsets. DAPT-v2 pretraining uses the unlabelled task paragraphs and applies to the encoders only, since Qwen3-32B is fine-tuned from its released checkpoint. Back-translated and synthetic paragraphs enter the task training sets only.

Component	Task 2 Closed	Task 2 Open	Task 1 Closed	Task 1 Open
Official test score	0.729	0.736	0.657	0.669
Editorial model	8 CAMeLBERT-mix taggers	8 CAMeLBERT-mix + 8 MARBERTv2	union of 3 Qwen3-32B seeds	as Closed
Debate model	8 MARBERTv2 taggers	8 CAMeLBERT-mix + 8 MARBERTv2	8 CAMeLBERT-mix + 8 MARBERTv2	8 CAMeLBERT-mix
Encoder pretraining	DAPT-v2, 1,042 paragraphs	as Closed	as Task 2	as Closed
Gold training data	train and dev, 829	as Closed	train and dev, 829	as Closed
Back-translation	none	none	global pass, 3× on ST and CO	as Closed
Synthetic data	none	158 span paragraphs	none	291 paragraphs
Cross-family blend	none	char-level, w_{CAM} 0.5 / 0.35	probability blend, debates	none
Thresholds	per domain and class	per domain and class, retuned	global, OOF recalibrated	per domain and class
Span compaction	G_{ed} 400, G_{db} 5, L 25	as Closed	not applicable	not applicable
Span fusion from Task 2	not applicable	not applicable	none	per-class rules, debates

Table 3: Exact components of the four final submissions. "As Closed" means identical to the Closed entry of the same task. The Task 1 Closed blend averages CAMeLBERT-mix and MARBERTv2 class probabilities on debates, and the Task 1 Open fusion reads labels off the Open Task 2 decode. Threshold values and seeds are in the text of this appendix.

differences we pursued on the leaderboard, which is why score distributions rather than single runs are the appropriate unit of comparison (Reimers and Gurevych, 2017). Averaging seeds removes that variability, with the standard deviation across ensembles falling from 0.030 at one seed to 0.025 at three, 0.016 at five, and zero at eight by construction. At a fixed threshold it does not raise the mean, which drifts from 0.513 to 0.488, because averaging shrinks the probabilities of rare classes and costs their recall. The value of ensembling here is stability rather than raw score, and converting that stability into score requires the per-class calibration our system applies on top.

Span compaction procedure. For each paragraph, and for each label independently, we collect the decoded spans carrying that label and sort them by start offset. We then scan them left to right, merging a span into its predecessor whenever the gap between them is at most G characters, so that the merged span runs from the first start offset to the last end offset. Finally we discard every surviving span shorter than L characters. The permitted gap is G_{ed} on editorials and G_{db} on debates, and no span is ever merged across labels.

Task 1 encoder. 8 epochs, learning rate 2e-5, batch 16, five to eight seeds, and robust median per-class thresholds. The development-phase thresholds (AS/AN/ST/TE/CO/OT) are [.425, .25, .2, .3, .125, .275] for Closed and [.425, .275, .275, .375, .125, .35] for Open. The evaluation phase used a global recalibration tuned out of fold by 5-

fold cross-validation over the 829 combined paragraphs, and thresholds were never tuned on test feedback. All per-system configurations are released with the code. Back-translation uses OPUS-MT `Helsinki-NLP/opus-mt-ar-en` and `en-ar`, with beam decoding for the global single pass and sampled decoding for the two extra rare-class (ST and CO) passes.

DAPT. 30-epoch masked language modelling with learning rate 5e-5, effective batch 16, warmup ratio 0.05, and weight decay 0.01. MARBERTv2 reaches that batch as 8 with gradient accumulation 2, which its larger vocabulary head requires. The development phase used the 829 train and dev paragraphs. The evaluation phase (DAPT-v2) used all 1,042 task paragraphs, namely train, dev, and the unlabelled test text, with labels never used, for both encoder families.

Qwen3-32B. 4-bit NF4 quantisation with double quantisation and bfloat16 compute. LoRA is applied to all attention and feed-forward projection matrices, with r=16, alpha 32, and dropout 0.05. Training uses learning rate 2e-4 with a cosine schedule and 3% warmup, batch size 1 with gradient accumulation of 8, three epochs, maximum length 896, and the loss on answer tokens only. The prompt contains the official annotation-guideline label definitions. We train three seeds, and the editorial predictions are the union of the seeds' label sets. On a controlled holdout with an identical split, where the encoder ensemble scores 0.715, the 14B model reaches 0.714 and the three 32B seeds reach 0.738, 0.767, and 0.728 standalone. The generative framing matters more than the parameter count, since an earlier backbone search fine-tuned AceGPT-v2-8B (Liang et al., 2024) with a classification head, which reached only 0.382 on a clean holdout, against 0.619 to 0.632 for CAMeLBERT-mix.

Three-seed union, checked after submission. The submitted union entered without a separate validation. We recomputed it on the controlled holdout above, routing editorials to the LLM and debates to the encoder ensemble as in the submission. One submitted seed's holdout predictions were not retained, so the check uses the other two submitted seeds and two further seeds from the campaign. In all four three-seed combinations the union beats the mean of its own seeds, by 0.007 to 0.018, and also beats majority voting and intersection of the same seeds (Table 4). The recall-oriented rule is therefore the best of the three at this seed count. The three combinations containing the best seed score 0.760 to 0.763, and the one without it 0.743. On test the union added 0.017 to 0.022 over a one-seed routed submission, a gain of the same order as the holdout margins above, although the two baselines differ.

Editorial rule	Min	Max
Encoder ensemble (no LLM)	0.715	0.715
Single Qwen3-32B seed (four)	0.729	0.768
Union of three seeds	**0.743**	**0.763**
Majority vote of three seeds	0.735	0.759
Intersection of three seeds	0.725	0.737

Table 4: Post-submission check of the three-seed union on the controlled Task 1 holdout (206 paragraphs), macro-F1 over the four seeds or the four three-seed combinations. The single-seed mean is 0.744, and within each combination the union exceeds the mean of its three seeds. Every row keeps the encoder ensemble on the debates and applies the stated rule to the editorials.

Qwen3 fine-tuning prompt (translated from Arabic). *"Classify the following paragraph of type (editorial / debate) by the argumentative evidence types it contains. Definitions: AS (assumption): assumptions, inferences, opinions, judgements, or claims that need support. AN (anecdote): evidence through personal experience, a story, a real event, or a concrete example. ST (statistics): quantitative evidence, studies, percentages, and figures, where a source need not be cited. TE (testimony): quoting or citing experts, bodies, or specific sources. In debates, restating the opposing team's claims counts as testimony. CO (common ground): agreed-upon knowledge, self-evident facts, or an objective explanation of how a procedure works. OT (other): does not genuinely contribute to the argumentative discourse (greetings, organisation, transitions). Output only the comma-separated list of codes present, with no explanation. Paragraph: {text}"* The Arabic original and the synthetic-data generation prompt templates ship with the code.

B Synthetic Data Details

Generation protocol. The generator is a proprietary large language model whose vendor and version we do not disclose. It was run in batches through the vendor's interactive interface, with its default sampling settings and no temperature or length parameter set. Each batch prompt con-

tained the official label definitions, three to five real training paragraphs of the target classes with their gold labels (or their labelled spans for Task 2), a topic and register assignment, and the output format. Paragraphs were validated against the official label definitions and deduplicated, with offsets checked programmatically. For Task 1 we targeted ST, CO, and OT and obtained 291 validated unique paragraphs (171 in the development phase). For Task 2 the paragraphs were authored as ordered labelled segments joined by single spaces, which gives exact offsets by construction. We obtained 158 paragraphs with 772 spans (88 in the development phase), with the rare types boosted (ST 128 against 38 real and CO 99 against 50). All synthetic data are used in training only and are released with the code, documented as machine-generated. The statistics inside synthetic ST spans are plausible but invented, and the recipe emits ordered, non-overlapping spans, so it cannot express nested annotations such as that in Figure 6.

Check by a second model. After our final submission, a second proprietary large language model, from the same family as the generator but a different model, re-annotated all 291 Task 1 paragraphs, shown only the official label definitions and blind to the generator's labels. Being same-family, it bounds annotation quality from above rather than validating it independently. Exact label-set agreement is 79.0%, micro-F1 is 0.94, and per-label Cohen's κ ranges from 0.61 (TE) to 1.00 (ST), with near-perfect agreement on the targeted rare classes (ST 1.00, CO 0.99). The blind annotations and the agreement script are released with the code.

Check by a human annotator. For the camera-ready version the first author, a native speaker familiar with the annotation guidelines, labelled 30 Task 1 synthetic paragraphs blind to the generated labels, 15 per genre and sampled to cover the targeted classes. Each was also rated for naturalness on a three-point scale. Table 5 gives the result beside the second model's annotations of the same 30 paragraphs. Agreement with the generator is micro-F1 0.84, with the targeted classes near-perfect (ST κ 1.00, CO 0.93). The human confirmed 96% of the generator's labels while only 74% of the human's labels were in the generator's set, so the disagreements are almost all labels the generator omitted rather than labels it invented. Fourteen of the fifteen disagreements add a label, most often TE for a restated opposing claim or OT for an opening address in a debate turn, the two debate-specific rules of the guidelines. Naturalness averaged 2.93 of 3, and no paragraph was rated unnatural. On the same paragraphs the second model agrees with the generator more closely than the human does (0.89 against 0.84), the direction the same-family caveat predicts. The sheet, the key, and the scoring script are released with the code.

	Human	**Model**
Exact label-set agreement	0.50	0.63
Micro-F1 against generator	0.84	0.89
Generator labels confirmed	0.96	0.92
Own labels in generator set	0.74	0.85
κ ST / CO	1.00 / 0.93	1.00 / 0.93
κ AS / OT	0.67 / 0.66	0.87 / 0.77
Naturalness (1 to 3)	2.93	

Table 5: Blind checks of the Task 1 synthetic labels on the same 30 paragraphs, by the first author and by the second model. TE and AN have three and one generated instances in this sample, so their κ values are not informative and are omitted.

Figure 3 shows an example synthetic Task 2 editorial paragraph authored as four ordered segments, translated here from the Arabic, whose offsets are those of the Arabic original, namely CO *"No one denies that e-commerce has radically changed shopping habits"* [0,60), AS *"yet the forgotten side is the environmental cost of ever-growing delivery trips"* [61,124), ST *"last-mile delivery emissions rise by about 30% annually in congested cities"* [127,195), and AS *"organising smart distribution hubs near neighbourhoods could tangibly ease this burden"* [196,266). Joining the segments with single spaces, with an unlabelled connective (*wa*, "and") allowed between spans, yields exactly these character offsets. A Task 1 synthetic CO example in the debate register reads *"No one denies that the Earth orbits the Sun in a near-elliptical orbit, and that together with the planet's axial tilt this produces the succession of seasons. This is established astronomical knowledge that every pupil studies, beyond dispute in a climate debate."*

C Threshold Recalibration

Conditions on the controlled result. Two conditions bound the experiment reported above. The comparison baseline is the development-phase threshold set, which was itself tuned on 612 of the 829 paragraphs, so it is not blind to most of each held-out split. Under a fully blind protocol, tuning on those 612 and evaluating on the untouched



Figure 3: The synthetic Task 2 paragraph described above (CO is common ground, AS assumption, and ST statistics, as in Figure 1). The offsets are exact by segment concatenation.

217, the loss is 0.006 against the 0.014 that 621 tuning paragraphs give in Table 7. That is a single split, and the split-to-split standard deviation at that tuning size is 0.015, so the reduction is smaller but not separable from noise. And the experiment concerns recalibration of an existing threshold set rather than the use of per-class thresholds, which every submitted configuration applies.

Relation to known results. The mechanism is not new. Fitting a decision rule to validation output and then reporting the tuned score is a form of selection bias, whose consequences for model selection and for the repeated reuse of a holdout are established (Cawley and Talbot, 2010; Dwork et al., 2015). What we add is a measurement of its size for one practice that remains common in shared-task pipelines. Within the same campaign, changes to the model itself were overstated by considerably less, although that comparison is observational.

Why the two groups might separate. Tuning-set size does not by itself separate them, since the thresholds and fusion rules were fitted out of fold on the same 829 paragraphs. The candidate explanation we favour is the number of free parameters a rule fits relative to the evidence available to fix them. A per-class threshold vector sets six parameters against a macro-F1 whose rare-class terms rest on nine and twelve development instances, and the per-label rules carry more, whereas a data or encoder change is a single decision judged against the whole score. We did not test this, and say so. A sweep from a single global threshold to per-class and then per-class-per-domain thresholds under the protocol above would test it directly.

The global recalibration entry. Global recalibration on Task 1 appears in Table 6 with a realised gain of 2.5 points, which the controlled experiment would not predict. No split at 207, 414, or 621 tuning paragraphs reaches that value, and only 0.4% of the splits at 746 paragraphs do. The two are not the same operation. The thresholds in the deployed configuration were the development-phase values, which had become stale against the evaluation-phase encoders, and the number of paragraphs receiving no label at all had risen from 16 to 22 of 213. That intervention repaired a threshold set that had ceased to match its model, rather than re-tuning one that already matched it. We record it as such, and note that on the evidence available we cannot exclude the alternative reading that it is a single favourable draw.

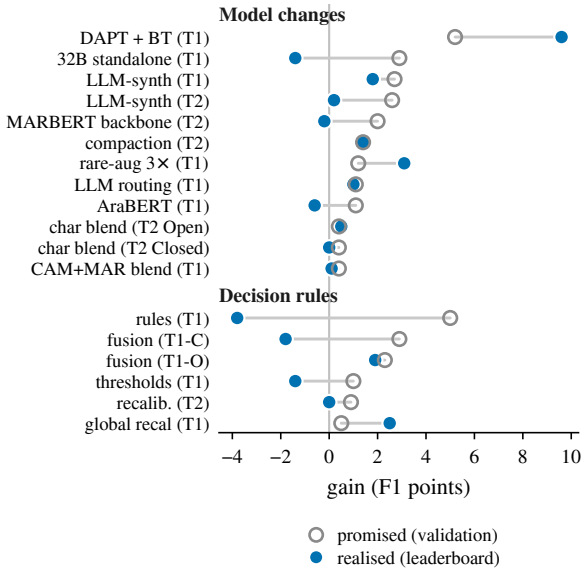


Figure 4: Promised validation gain (open) and realised leaderboard gain (filled) for the eighteen interventions, in F1 points, sorted by promise. Labels are abbreviated forms of the row names in Table 6, where the full data appear.

Notes on Table 6. The validation column is a pooled gain except for the Task 2 threshold recalibration row, whose entry averages the two per-domain gains of 1.2 and 0.6 points. The table lists the interventions for which we recorded both a validation estimate and a leaderboard delta. It was assembled from our own log after the fact, so it is necessarily incomplete, and an eightfold augmentation variant, class weighting, and domain routing on Task 1 were scored without a paired estimate and are omitted. Some realised deltas carry a co-occurring change. The MARBERTv2 backbone row also varied ensemble size, the Task 1 LLM-

Intervention	Val. basis	Val.	Real.
Model changes			
DAPT + BT (T1)	holdout	+5.2	+9.6
rare-aug 3× (T1)	holdout	+1.2	+3.1
LLM-synth (T1)	holdout	+2.7	+1.8
LLM-synth (T2)	holdout	+2.6	+0.2
span compaction (T2)‡	dev-tuned	+1.4	+1.4
char-level blend (T2 Open)	OOF	+0.4	+0.5
char-level blend (T2 Closed)	OOF	+0.4	0.0
LLM routing, one seed (T1)	holdout	+1.1	+1.0
AraBERTv2 in ensemble (T1)	holdout	+1.1	−0.6
32B standalone (T1)	holdout	+2.9	−1.4
CAM+MAR blend (T1)	OOF	+0.4	+0.1
MARBERTv2 backbone (T2)	CV	+2.0	−0.2
Decision-rule changes			
global recalib. (T1)	OOF	+0.5	+2.5
per-domain thresholds (T1)	OOF	+1.0	−1.4
span fusion (T1 Closed)†	OOF	+2.9	−1.8
span fusion (T1 Open)†	OOF	+2.3	+1.9
threshold recalib. (T2)	OOF	+0.9	0.0
per-label rules (T1)	holdout	+5.0	−3.8

Table 6: Data behind Figure 4, in F1 points. Val. is the gain on the stated validation basis, where OOF means out of fold, and Real. the gain on the official leaderboard, development or test according to the phase. Rows marked † entered together with the per-domain thresholds above them, and the row marked ‡ audits the first compaction grid. Notes on individual rows follow in the text.

synth row also carried a threshold retune, and both Task 2 char-level blend rows entered with a per-domain threshold retune. The blend was submitted on both Task 2 tracks from one validation estimate, gaining 0.5 on the Open board and nothing on the Closed one, where two distinct systems had already reached the same score. The two fusion rows entered together with the per-domain thresholds, so their isolated effect is smaller, and an isolating probe puts Closed fusion at −0.4. The compaction row audits the first grid, while the deployed configuration was worth 0.017 in total. DAPT and back-translation are bundled because they entered one submission, with separate holdout gains of +1.2 and +4.0. The routed-LLM holdout used the 14B model, and its realised entry averages the Closed and Open submissions (+0.8 and +1.1). The three-seed union is absent because it entered without a validation estimate, and its post-submission check is in Table 4. Validation estimates and realised gains are not usefully correlated within either group at this sample size, and each Pearson coefficient is set by one high-leverage intervention, so we do not rely on them. Compaction sits in the model group, although its dev-tuned gap thresholds would equally justify placing it among the decision rules.

Threshold recalibration under a controlled split. The leaderboard evidence in Figure 4 is observational, so we repeated the exercise under a controlled protocol, as a post-submission analysis, on the out-of-fold predictions of the Task 1 encoder ensemble, whose thresholds we recalibrated. For each of 500 random splits of the 829 paragraphs, rather than one fixed split (Gorman and Bedrick, 2019), we recalibrated the per-class thresholds on one part by an independent per-label grid search against the development-phase set, then measured the change in macro-F1 on both parts. Table 7 gives the result. Recalibrating on 207 paragraphs, roughly the size of the development set, appears to gain 0.033 but loses 0.016 held out and is non-positive in 97% of splits. Enlarging the tuning set shrinks the apparent gain to 0.010 at 746 paragraphs while the held-out loss does not move, so what depends on the size of the tuning set is the apparent gain rather than the gain delivered. The expected loss is close to the 0.014 that per-domain thresholds cost us on test. That is consistent with recalibration behaving this way at this data size, though one test delta of that size cannot be separated from sampling noise at the spread we observe. We repeated the experiment on Task 2, over the out-of-fold span probabilities of the submitted Closed system, scored with the official overlap-F1 and recalibrated by the same coordinate ascent that system uses, per genre on that genre's own score. The baseline is the development-phase threshold set of the submitted decode (Appendix A), fitted a month earlier on pre-DAPT-v2 cross-validation probabilities and therefore blind to the predictions under test, though, as on Task 1, not to the underlying texts. Over 200 splits, tuning on 207 paragraphs appears to gain 0.014 but loses 0.007 held out and is non-positive in 98% of splits. At 746 the apparent gain falls to 0.005 with the loss unchanged, and at an even split the figures are +0.008 in sample against −0.005 held out, non-positive in 92%. Task 2 therefore reproduces the Task 1 shape at roughly two fifths of the magnitude, so the effect is not particular to macro-F1 or to paragraph labels. Its size depends on the protocol, since ascending on the pooled objective rather than per genre halves the held-out loss to 0.003 and scoring the genre mean gives 0.002, but it is negative on average at every tuning-set size under every variant we tried. The held-out standard deviation is 0.004 to 0.008 and the splits are draws from one set of

829 paragraphs, so we attach no significance test, and the loss is carried mainly by the editorials. Our reimplementation reproduces the per-genre scores logged for that configuration, 0.729 on editorials and 0.733 on debates.

Tuned on	In-sample	Held-out	Held-out ≤ 0
207 paragraphs	+0.033	−0.016	97%
414 paragraphs	+0.019	−0.014	93%
621 paragraphs	+0.013	−0.014	87%
746 paragraphs	+0.010	−0.015	77%

Table 7: Threshold recalibration measured on the tuning subset and on the held-out remainder, averaged over 500 random splits of the 829 out-of-fold Task 1 paragraphs. The apparent gain shrinks with the tuning set while the held-out loss does not.

D Error Taxonomy

This appendix reports post-submission analyses on the released dev gold. Table 8 classifies every span of a routed dev decode, the same one used for the paired compaction bootstrap and weaker than the submitted routed system, before and after compaction. A gold span is *covered* when at least 80% of its characters receive same-label credit, *partial* when some do, and *missed* otherwise. A predicted span is *clean* when at least 80% of its characters earn same-label credit, *partial* when some but fewer than 80% do, a *label confusion* when it mostly overlaps gold spans of another label, and *hallucinated* when it earns no same-label credit and covers gold spans of other labels over less than half its length.

	Raw	Compacted
Gold spans (961)		
covered (≥80%)	642	656
partially covered	140	96
missed	179	209
Predicted spans		
total	1,167	605
clean (≥80%)	731	379
partial	82	105
label confusion	319	115
hallucinated	35	6
pred. spans per matched gold	1.20	1.02

Table 8: Task 2 error taxonomy on dev for the routed dev decode, before and after span compaction. The last row averages, over the gold spans that received any same-label credit, the number of predicted spans overlapping each one. It is not the ratio of the two totals above it.

We make three observations. First, compaction removes fragmentation almost entirely, since each matched gold span is covered by 1.02 predicted spans on average (1.20 raw) and hallucinated spans fall from 35 to 6, largely through the minimum-length filter. Second, the trade-off is explicit, as 30 further short gold spans are missed after filtering, a cost the pooled metric more than offsets. Third, the residual error mass is semantic rather than segmental, with 115 label-confusion spans and 209 whole-span misses. That is consistent with our later gains coming from data and models rather than from further decoding rules, although this taxonomy is post-submission and did not inform those choices.

Label	Support	FN	FP	Predicted instead
AS	157	6	26	TE (2)
AN	68	24	24	OT (7), AS (5)
ST	9	2	1	none
TE	76	13	11	OT (4), AN (2), AS (2)
CO	12	10	12	AN (4), AS (2)
OT	63	12	38	none

Table 9: Task 1 confusion on dev for the final Closed development system. FN and FP are per-label false negatives and positives. The last column lists every label that was predicted, but not gold, on at least two of the paragraphs where the gold label was missed.

Table 9 gives the Task 1 picture. CO fails by omission, with 10 of its 12 gold paragraphs missed and typically labelled AN or AS instead, exactly the confusion of Figure 9. ST, the target of our augmentation, is mostly solved (2 of 9 missed). The bulk of the errors sits on the boundary between AN and OT (7 of 24 missed AN paragraphs were called OT, and OT draws 38 false positives), and AS is essentially saturated. Rare-class semantics, not thresholding, is what remains.

Why editorials are harder. Every ranked team entry on the four leaderboards scored lower on editorials than on debates. The one reversal on the sheet is an unranked training-subset slot, ttlab's editorials-only Task 2 Closed variant at 0.623 against 0.601. Table 10 locates the cause in the training data rather than in any one system. The deficit is concentrated in two labels. OT is effectively a debate category, since only 5% of its training spans are editorial, and editorial OT is not recovered at all by this decode (F1 0.000 against 0.602 on debates). AS, the majority class in both genres, draws 71% of its training spans from debates. The labels whose training spans are mostly

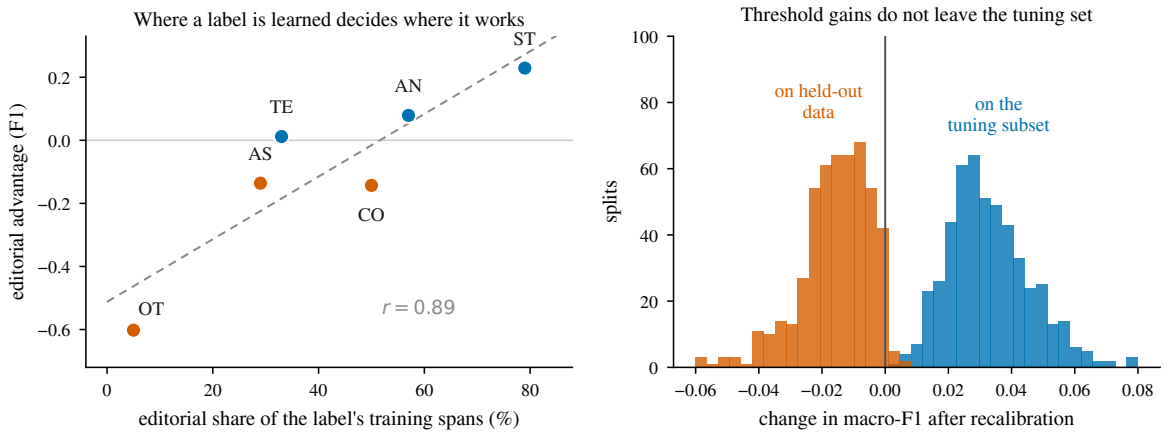


Figure 5: The two post-submission findings. Left, each label's editorial advantage on the development set against the editorial share of that label's training spans, with OT at the extreme and the dashed line a least-squares fit. Right, the change in Task 1 macro-F1 after recalibrating per-class thresholds on a quarter of the out-of-fold paragraphs, measured on the tuning subset and on the held-out remainder over 500 random splits. The in-sample distribution lies entirely above zero, while the held-out distribution is non-positive in 97% of splits.

editorial, ST and AN, are handled better on editorials, as is TE by a smaller margin. Across the six labels the editorial advantage tracks the editorial share of training spans, with a Pearson r of 0.89 over six points. The coefficient is stable under leave-one-out (0.78 to 0.95) but is anchored by the two extremes and is not significant without OT (p=0.117), so we read it as descriptive.

The pattern is not an artefact of our routing. It appears with the same strength in each encoder family alone (0.92 for CAMeLBERT-mix and 0.92 for MARBERTv2, against 0.88 for the routed decode, all before compaction), and on Task 1 the two extremes behave the same way, with OT at −0.34 and AN at +0.35. Span length is at most a secondary factor. Editorial gold spans are longer on average (99 against 76 characters), which costs more under the partial-overlap metric, but restricted to AS the difference is small (78.7 against 74.1). Editorials are also more label-diverse, with AS 55%, AN 20%, and TE 16% of gold spans against AS 63% and OT 16% in debates, so a system that fits the dominant debate pattern has less to fall back on.

Our synthetic data support this reading, as they were written with editorial cover for the rarest classes. The Task 1 systems with and without them differ in the synthetic paragraphs and the threshold recalibration these induce. The editorial column rises for the classes that received editorial synthetic paragraphs while the debate column stays flat, by +0.152 against −0.008 for CO and +0.075 against 0.000 for ST, on five and eight editorial dev paragraphs. The exception is OT, where the 30 editorial OT paragraphs of the development-phase pool did not help (−0.028), which suggests that a residual category defined by what it is not is harder to synthesise than a contentful one. Genre-balanced annotation of the rarest cells, rather than more modelling, is what we would expect to close this gap.

Label	Ed F1	n	Db F1	n	Diff	% ed
AS	0.655	144	0.791	443	−0.136	29
AN	0.557	51	0.478	60	+0.079	57
ST	0.652	8	0.423	1	+0.229	79
TE	0.772	41	0.760	71	+0.012	33
CO	0.000	5	0.143	11	−0.143	50
OT	0.000	12	0.602	114	−0.602	5

Table 10: Task 2 dev F1 by genre and label for the routed dev decode after compaction, against the editorial share of that label's training spans. The two labels drawing the smallest share of their training spans from editorials, OT and AS, are also the two on which the editorial deficit is concentrated. n is the gold span count for that label in that genre. Several cells rest on single-digit support, so individual F1 values there are unstable even where the pattern across labels is not.

A controlled genre-transfer test. The evidence above is correlational, so we ran a size-matched training experiment. We trained the Task 1 encoder on 255 editorial paragraphs, on 255 debate paragraphs, and on a half-and-half mixture of the same size, with thresholds fixed at 0.5 and three seeds each. We evaluated on the editorial and debate halves of the development set. These are deliberately small single models, not competitive configurations, so the absolute values sit far below

those of our submitted system, and only the contrast between them is meaningful. Table 11 gives the means. The crossover is clean and holds in all six seed comparisons, since training on editorials gives 0.350 on editorials against 0.271 on debates, while training on debates reverses this to 0.207 and 0.395. With architecture held fixed, genre composition alone moves a model's strength from one genre to the other, and the half-and-half mixture recovers nearly all of both columns. This does not show that architecture is irrelevant, only that composition suffices to produce the split, since the routing result of Section 3.1 shows two encoder families splitting by genre on identical data. One label does not follow this pattern. An editorials-only model scores 0.000 on OT at all three seeds and in both genres, because only 20 of the 222 OT training paragraphs are editorial (9%), so its 255-paragraph pool carries just 20 OT positives, whereas a debates-only model reaches 0.374 on editorial OT. Genre balance therefore helps only where the target genre exemplifies the label. Where the cell is thin, as it is for editorial OT, the remedy has to be annotation rather than rebalancing.

Trained on	Ed	Db
editorials only	**0.350** (0.038)	0.271 (0.036)
debates only	0.207 (0.008)	**0.395** (0.012)
half and half	0.319 (0.035)	0.390 (0.016)

Table 11: Size-matched genre-transfer test on Task 1 (255 training paragraphs each), dev macro-F1 by genre, averaged over three seeds with thresholds fixed at 0.5, and the standard deviation over seeds in brackets.

E Suggestions for Future Editions

The task is a valuable new resource, and we offer four observations for future editions in a constructive spirit. The cross-team figures below are post-submission, from the organisers' published sheet.

Report uncertainty. Task 1 macro-F1 at this data size carries a development-set resampling half-width of about ± 0.05. A bootstrap interval beside each leaderboard score, or larger rare-class test support, would make small ranking gaps interpretable.

Revisit the span scorer. The overlap scorer rewards merging essentially all same-label spans within an editorial paragraph (Appendix A). An IoU-style or length-normalised variant would remove that incentive.

Limit leaderboard feedback. With unlimited submissions the leaderboard doubles as a validation signal for every team, ourselves included. A small frozen holdout or a cap on scored submissions would sharpen the comparison.

Weight the genres. Debates supply 73% of the gold spans on the development set, and the test set has the same genre proportions, so the pooled Task 2 score is in effect a debate score. Across the eight Task 2 Closed teams the debate column correlates with the overall score at 0.96, while the editorial column, the harder part, correlates at 0.42 and barely affects rank. Among the top five on Task 1 the balance reverses and the editorial column is the separating one. Averaging the two domain scores, or reporting the per-domain columns as the primary result, would let editorial progress count towards the final standing.

F Qualitative Examples

All examples are from the released development set, and the translations are ours.

Nested gold spans. Figure 6 shows the development set's one *nested* annotation (editorial 962), where an ST span is embedded inside a sentence-length AN span. BIO-style tagging cannot represent this, while per-class sigmoid tagging can. The paragraph translates as *"Even if you are among those who read the small print before accepting privacy settings, which 56% of Americans do not do according to a 2023 study, some sites still request permission to share information about your visit."*

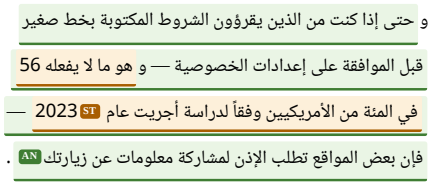


Figure 6: Nested gold annotation (editorial 962). An ST (statistics) span sits inside an AN (anecdote) span, whose green underline marks its extent. Our multi-label sigmoid tagger can predict both, while a BIO scheme cannot.

Why domain routing works. Figure 7 contrasts the two encoder families on debate turn 390, which is in a fully oral register. Both gold TE spans restate the opposing speaker's claims, which count as testimony under the guidelines' debate-specific rule.

CAMeLBERT-mix fragments the spans into pieces as short as three characters and drifts towards OT, while MARBERTv2 recovers both TE spans almost exactly and returns the two gold AS spans as one (paragraph overlap-F1 0.94 against 0.42, at full recall and precision 0.88). The turn translates as *"He said the pillars of the state are people, land, and government, and kept repeating that in the affected countries the land has been wiped away. Let him look at the nearest affected country while forgetting the state's elements in first-world countries, and ask yourself how this resolution affects the land pillar there."*



Figure 7: Gold against CAMeLBERT-mix and MARBERTv2 on debate turn 390 (TE is testimony, AS assumption, and OT other). CAMeLBERT-mix fragments and mislabels the spans, while the dialect-pretrained MARBERTv2 recovers the testimony spans and their extents, merging the two assumption spans into one. This is the complementarity behind our domain routing.

Span fragmentation (Task 2). Consider debate paragraph 186 (Figure 8). The gold AS span covers characters 154 to 316 and reads *"We came today as the proposition team, strongly in favour of the motion, namely that this house would prevent for-profit companies from using social and political causes in marketing."* The raw MARBERTv2 decode emits nine AS fragments in this paragraph, of which [185,197), [200,204), and [211,316) fall inside the gold span and [319,322) falls just past it. Token probabilities dip below threshold on short function words, leaving gaps of three to seven characters, and two fragments are under ten characters. Under the overlap metric each fragment is charged separately to precision. Gap merging joins fragments separated by at most five characters, but with $G_{\mathrm{db}}{=}5$ the chain continues past the gold boundary to [211,677). Compaction therefore removes the fragmentation at the cost of over-running the span, the trade-off that the plateau in Appendix A implies.



Figure 8: Fragmentation on debate paragraph 186. The gold AS (assumption) span (top) against the raw token-threshold decode (bottom), which splits it into fragments at short function words. Span compaction with $G_{\mathrm{db}}{=}5$ joins the last two fragments into a chain that runs to character 677, well past the gold boundary, while the first two merge into a 19-character span that the minimum-length filter then discards. Neither outcome is shown in the figure.

Rare-class confusion (Task 1). Editorial paragraph 95 (Figure 9), with gold {CO} and prediction {AS}, reads *"These tools include private web browsers, encrypted messaging apps, password managers, tracker blockers, and virtual private networks (VPNs)."* An objective, uncontested enumeration, which is common ground under the guidelines, is misread as an assumption. With only 36 CO training paragraphs, this boundary must be calibrated from a handful of examples, which is where our targeted augmentation was aimed.

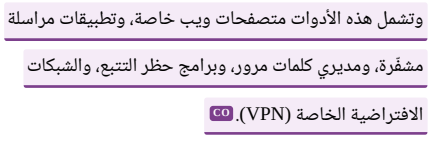


Figure 9: Editorial paragraph 95, an objective enumeration of privacy tools. The gold label is common ground (CO) and the prediction was assumption (AS).